

# Explainable Deep Learning for Cyber Attack Detection in Electric Vehicle Charging Stations

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## Abstract

The growing integration of Electric Vehicles (EVs) into smart grids has led to significant cybersecurity challenges, particularly concerning vulnerabilities in Electric Vehicle Charging Systems (EVCS). These systems, which facilitate secure interactions among EVs, EVCS, and management systems through protocols such as ISO 15118 and OCPP, are increasingly exposed to cyber threats due to the rise of Internet of Things (IoT) connectivity. Threats such as denial of service, information gathering, man-in-the-middle, and injection attacks pose serious risks to the stability of charging stations and the overall infrastructure. This paper introduces a novel Deep Learning model to cyber attack detection utilizing the CIC EV Charger Attack Dataset 2024 (CICEVSE2024). Although the original study in this dataset reported an accuracy of 78.87%, our proposed deep learning model achieves a remarkable accuracy of 97.15%, with a precision of 97.19%, recall of 97.15%, F1 score of 97.15% and an area under the curve (AUC) of 97.14%. These results demonstrate the effectiveness of advanced deep learning techniques in enhancing the security of the EV charging infrastructure. The Shap explainable AI results revealed the significant impact of 12 selected features on detecting benign and attack data using our deep learning model.

## CCS Concepts

• Security and privacy → Network security; Systems security.

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## Keywords

Electric Vehicle Charging Station (EVCS), Attack Detection, Explainable Deep Learning, Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM)

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## 1 Introduction

The shift towards electric transportation is essential for addressing climate change and mitigating greenhouse gas emissions, with Electric Vehicles (EVs) being pivotal in this transition [13]. Nevertheless, the extensive adoption of EVs brings forth new cyber-physical security challenges, as these vehicles can be manipulated to breach and disrupt energy networks, thereby threatening both the stability of infrastructure and the safety of EV users [1]. The increasing momentum for EVs is fueled by their environmental and economic advantages, which further hastens the phase-out of fossil-fuel-powered vehicles [6].

The incorporation of EVs into Smart Grids (SGs) enhances energy efficiency and sustainability, yet it simultaneously heightens vulnerability to cybersecurity threats. While SGs facilitate energy management through bidirectional communication [11], they are increasingly susceptible to cyberattacks, especially as EVs emerge as potential conduits for such threats, endangering grid stability [21]. As the adoption of EVs accelerates, it becomes imperative to secure the interconnected framework of these vehicles to avert attacks that could jeopardize both personal safety and energy infrastructure [2].

EVs function as interconnected entities within cyber-physical systems, which introduces distinct security challenges. Despite their rising significance, the field of EV cybersecurity remains relatively under-researched, with a growing awareness of potential cyber threats underscoring the necessity for enhanced protective measures. These threats not only endanger individual safety and

privacy but also pose significant risks to the overall energy grid, raising alarms about the sustainability of widespread EV adoption [1].

The emergence of Electric Vehicle Charging Stations (EVCSs) is vital for supporting the expansion of EVs, providing crucial infrastructure to satisfy consumer needs [12]. These stations act as critical interfaces between SGs and EVs, integrating several essential components. EVCS integrates several key components including a Charging Station Management System (CSMS), one or more Charging Stations (CS), and Electric Vehicle Supply Equipment (EVSE) to facilitate vehicle charging. Figure 1 demonstrates the network architecture of EVCS.

- **Electric Vehicle Supply Equipment (EVSE):** The EVSE functions as the essential interface for electric vehicles (EVs), enabling the transfer of charging and communication information between the vehicle and the charging infrastructure, in accordance with the ISO 15118 standard.
- **Charging Station (CS):** The CS manages multiple charging units, each equipped with an EVSE and power connectors for electric vehicles. It is governed by the CSMS, which oversees the charging operations and enforces power limitations. Standards such as ISO 15118 and OCPP enhance the functionality of the station by facilitating user authentication, transaction management, and billing processes.
- **Charging Station Management System (CSMS):** The CSMS acts as the central controller, facilitating communication between the CS and the EVSE. It determines service parameters based on user preferences, vehicle conditions, and grid status. Furthermore, the CSMS handles data management, user applications, and billing services, while also incorporating smart charging protocols.
- **Distribution System Operator (DSO):** The DSO functions as a supervisory entity, monitoring the distribution of electricity to maintain grid stability. It gathers and analyzes data from various CSMS to enhance overall operational efficiency.

However, the integration of physical components like plugs and chargers with cyber technologies such as payment terminals and communication protocols exposes EVCSs to significant cybersecurity risks. As these systems rely on continuous Internet connectivity and complex communication with the grid, they become vulnerable to cyberattacks, which could disrupt both individual users and the broader energy network [9]. Common threats in EVCSs include Denial of Service (DoS) attacks, information gathering techniques, Man-in-the-Middle (MitM) attacks, and injection attacks.

The implementation of Anomaly Detection Systems (ADS) is becoming an efficient approach to address these risks [24]. Unfortunately, there is a lack of detection methods for EVCSs. In this paper, we present a comprehensive study on Deep Learning (DL)-based attack detection in EVCSs.

In the following section, we discuss existing research and its shortcomings. Section 3 outlines our detailed methodology for data set selection and evaluation of the performance of DL algorithms. Following this, we discuss our results in Section 4. Finally, we conclude our research in Section 5.

## 2 Related Work

The swift expansion of EVs has underscored the necessity for comprehensive cybersecurity strategies within EVCS. Although various studies have explored methods to safeguard these essential infrastructures, many of them utilize generic Internet of Things (IoT) datasets that fail to adequately reflect the unique attributes of EVCS environments [25].

To mitigate these vulnerabilities, the authors in [4] developed a DL-based intrusion detection system (IDS) that employs Deep Neural Network (DNN) [22] and Long Short-Term memory (LSTM) [14] algorithms to identify DoS attacks targeting EVCS. Both approaches demonstrated high levels of accuracy, and LSTM exceeded DNN in terms of precision and recall. However, the research was limited to DoS attacks, limiting its applicability to a wider spectrum of cybersecurity threats. In addition, the computational demands of deep learning techniques, particularly LSTM, present challenges for real-time implementation. The reliance on the CICIDS 2018 dataset, which is not tailored to IoT contexts, further restricts its relevance to the specific cybersecurity issues faced by EVCS.

In a subsequent investigation, the authors in [5] introduced a DL framework for ransomware detection in EVCS, evaluating the efficacy of DNN, Convolutional Neural Networks (CNN) [17], and LSTM algorithms. DNN was noted for its quicker training times and lower complexity, while LSTM achieved the least variance and the highest Area Under the Curve (AUC). Despite these advancements, there remains considerable opportunity to enhance detection rates and overall accuracy, especially in the context of ransomware threats.

In [19], the authors proposed an anomaly detection framework for the power supply systems of EV charging stations, utilizing a multi-head attention model that emphasizes network traffic headers, payloads, and message sequences. Although this model showed potential in enhancing detection accuracy, it faced difficulties in effectively identifying anomalies within high-dimensional network traffic, a prevalent issue in EVCS networks characterized by intricate interactions.

The research presented in [23] examined security issues within EVCS, with a particular emphasis on the Open Charge Point Protocol (OCPP). The authors devised a backpropagation neural network aimed at distinguishing between legitimate and malicious traffic, while also tackling the challenges and expenses associated with the implementation of Transport Layer Security (TLS) in EVCS. Despite offering significant insights into real-time security monitoring, the study was constrained by its concentration on vulnerabilities specific to the protocol and the substantial costs involved in deploying comprehensive security measures like TLS.

In [29], the authors introduced a streamlined authentication framework for EVCS networks that utilizes nonsupersingular elliptic curves to establish a lightweight protocol for communication between EVs and the grid. Nonetheless, the dependence on nonsupersingular elliptic curves raises questions regarding the security and efficiency of the scheme, given the intricate structures of point groups and the potential for vulnerabilities to various attacks.

The investigation conducted in [28] assessed a range of machine learning methodologies, including Random Forest (RF), Naive Bayes, Multi-layer Perceptron, Support Vector Machine, and AdaBoost,

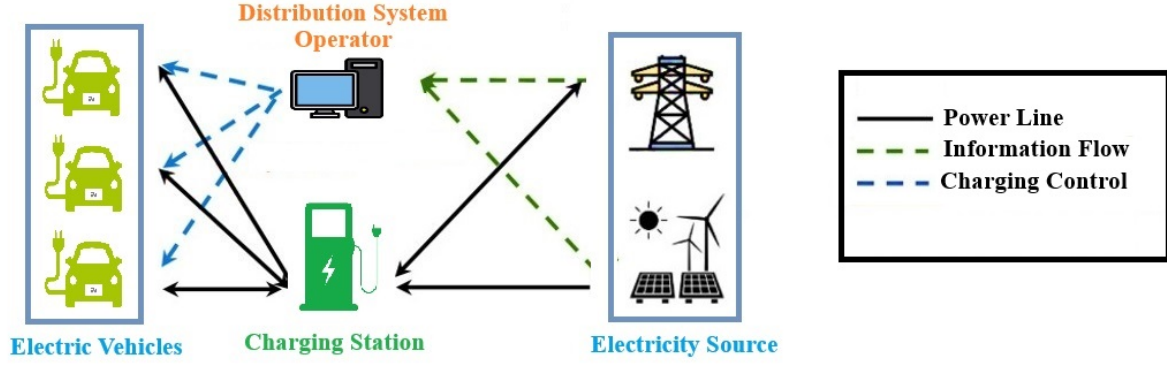


Figure 1: Network Architecture for EVCS

utilizing the IoT-23 dataset. While RF yielded the highest accuracy, the reliance on a generic IoT dataset restricts the relevance of the findings to the specific challenges faced by EVCS.

In [8], the authors proposed a Federated Learning (FL) framework aimed at detecting anomalies within IoT networks, incorporating differential privacy to bolster user confidentiality and utilizing blockchain technology to safeguard FL algorithms against model poisoning attacks. Although this methodology illustrates the potential for integrating sophisticated techniques such as CNN-LSTM into the security of EVCS, the intricacies associated with privacy-preserving technologies like blockchain present challenges related to scalability and resource management.

The researchers in [24] employed the CICEVSE2024 dataset to develop a privacy-preserving anomaly detection framework for Electric Vehicle Charging Stations (EVCS) utilizing Federated Learning (FL). They reported achieving high accuracy in identifying cyber attacks targeting EVCS. Nonetheless, the study lacked comprehensive details regarding the specific models implemented for cyber attack detection, and the application of the same dataset for both training and testing raises significant concerns regarding overfitting, which constitutes a notable limitation of this research.

Although current studies provide important perspectives on the security of EVCS, there is a pressing need for more specialized strategies that specifically address the different cybersecurity issues faced by EVCS. Notably, deep learning techniques necessitate further enhancement to ensure their effectiveness in real-time scenarios, scalability, and comprehensive threat management.

### 3 Methodology

Our objective is to assess the efficacy of DL algorithms in identifying cyber threats within smart home networks. To achieve this, we propose a methodology that involves the selection of an appropriate dataset and the subsequent evaluation of the performance of DL algorithms.

#### 3.1 Dataset Selection

For our investigation, we have chosen to collect a dataset from real-implemented testbeds. These testbeds provide data that reflects actual conditions and behaviors in real-world environments, ensuring the relevance and applicability of methodologies and models [27].

Based on this criteria, we have selected the CIC EV Charger Attack Dataset (CICEVSE) 2024, which is specifically designed for examining and identifying cyber threats targeting EVCS [7]. The dataset was generated using a testbed setup involving an operational Level 2 charging station, EVSE-A, alongside various Raspberry Pi devices implementing the Electric Vehicle Communication Controller (EVCC), EVSE-B, Power Monitor, and the local CSMS. The EVSE-A communicates with the remote CSMS platform via OCPP protocol, while EVSE-B uses ISO15118 and OCPP protocols to communicate with the EVCC and local CSMS. The power consumption of EVSE-B is monitored through an I2C wattmeter.

The dataset contains benign and attack scenarios that target the EVSE during both idle and charging states. The attack scenarios include network and host-based attacks such as reconnaissance, DoS, backdoor, and cryptojacking. Both datasets have benign, reconnaissance, and DoS data. Additionally, dataset EVSE-B includes extra backdoor and cryptojacking data. For our research, we focused on these three classes and omitted the backdoor and cryptojacking data from EVSE-B.

This dataset is designed by implementing real testbed. Earlier literatures worked on IoT datasets which are not designed for EVCSs. The use of this dataset will enhance the applicability of our research in real EVCS networks.

#### 3.2 Evaluating the Performance of Deep Learning Algorithms

In this phase of analysis, we applied DL algorithms to evaluate the performance of detecting various cyberattacks targeting EVCS. Specifically, we utilized four DL models: DNN, CNN, LSTM, and a hybrid CNN-LSTM model, to evaluate their performance in identifying attacks.

Before applying the DL models, we removed unnecessary features, such as packet source and destination addresses, to both anonymize the data and mitigate the risk of overfitting. This also helped reduce computational overhead. To ensure the training and test sets maintained comparable statistical characteristics, we conducted a comparison of summary statistics, including mean, median, and variance. This step was crucial for ensuring the models' ability to generalize effectively. Additionally, the Kolmogorov-Smirnov

**Table 1: Selected features for attack detection**

| Sl | Feature Name               | Description                                                                                                |
|----|----------------------------|------------------------------------------------------------------------------------------------------------|
| 1  | src_port                   | The port number utilized by the source in the network connection.                                          |
| 2  | bidirectional_duration_ms  | The cumulative duration of the bidirectional data flow measured in milliseconds (ms).                      |
| 3  | src2dst_duration_ms        | The time taken for the data flow from the source to the destination, expressed in milliseconds (ms).       |
| 4  | dst2src_duration_ms        | The time taken for the data flow from the destination back to the source, expressed in milliseconds (ms).  |
| 5  | bidirectional_mean_piat_ms | The average Packet Inter Arrival Time (PIAT) in milliseconds for the entire bidirectional data flow.       |
| 6  | src2dst_min_piat_ms        | The minimum Packet Inter Arrival Time (PIAT) from the source to the destination, measured in milliseconds. |
| 7  | src2dst_mean_piat_ms       | The average Packet Inter Arrival Time (PIAT) from the source to the destination.                           |
| 8  | bidirectional_min_piat_ms  | The Minimum Packet Inter Arrival Time (PIAT) in milliseconds for the complete bidirectional flow.          |
| 9  | bidirectional_ack_packets  | The amount of acknowledgment packets within the two-way flow.                                              |
| 10 | dst2src_ack_packets        | Count of acknowledgment packets detected from the destination to the source.                               |
| 11 | dst2src_psh_packets        | Count of push (PSH) packets detected from the destination back to the source.                              |
| 12 | application_is_guessed     | Determines whether the active application is inferred from network activity patterns.                      |

test [18] was employed to further assess the similarity of the distributions between the two sets.

Then, we employed the SelectKBest algorithm [16] to select the top 10 most relevant features from the dataset. This feature selection procedure was instrumental in ensuring that only the most informative features were utilized for model training, thereby minimizing computational complexity and enhancing the models' ability to accurately detect anomalies in the EVCS environment.

Each dataset was partitioned into training (80%) and testing (20%) sets to ensure sufficient data for model evaluation while avoiding overfitting [10]. All numerical features were normalized using StandardScaler [26], which centers the data around the mean and scales it according to the standard deviation. The formula for StandardScaler is as follows:

$$X' = \frac{X - \mu}{\sigma} \quad (1)$$

In this equation,  $X$  represents the original value,  $\mu$  denotes the mean of the feature,  $\sigma$  signifies the standard deviation, and  $X'$  indicates the standardized value.

This normalization process guarantees that the features maintain a uniform range, thereby preventing the models from being influenced by variations in feature magnitudes. This consistency is essential for deep learning models, including CNN, LSTM, and hybrid architectures, to learn effectively from the provided input data.

We evaluated DL models using standard metrics: accuracy, precision, recall or True Positive Rate (TPR), F1 score, and Area Under the ROC Curve (AUC) [15], all defined in terms of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). These metrics are calculated using the following equations:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (2)$$

$$\text{Recall/TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (3)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (4)$$

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

$$\text{False Positive Rate (FPR)} = \frac{\text{FP}}{\text{FP} + \text{TN}} \quad (6)$$

$$\text{AUC} = \sum_{i=1}^n (\text{TPR}_i + \text{TPR}_{i-1}) \cdot (\text{FPR}_i - \text{FPR}_{i-1}) \quad (7)$$

These metrics offer a comprehensive evaluation of the ability of the models to detect cyberattacks in EVCS. By comparing the performance of DNN, CNN, LSTM, and CNN-LSTM models, we will identify the optimal configuration for real-time anomaly detection in EVCS.

Besides, Explainable AI (XAI) aims to make the decision-making processes of AI models transparent and interpretable, ensuring that users can understand how inputs are transformed into outputs [3]. This transparency is crucial for fostering trust and accountability in AI applications. In our research, we will apply SHAP [20] to explain our model's predictions, allowing us to identify the most influential features and gain deeper insights into the factors driving our results. This will enhance the interpretability of our model and support its effective and responsible use.

## 4 Results

After using the SelectKBest algorithm to select the 10 best features separately, we found different features for the EVSE-A and EVSE-B data, with 9 common features for both testbeds. Ultimately, we chose 12 unique features from Table 1 to use in our DL models.

After applying all four deep learning models with different configurations of DNN, CNN, LSTM, and a hybrid CNN-LSTM, we discovered that the hybrid CNN-LSTM achieved high accuracy in detecting attacks. The configuration or parameters used in our hybrid CNN-LSTM model are listed in the Table 2. This model combines the strengths of CNNs for extracting features from sequential data and the ability of LSTMs to capture temporal dependencies. This hybrid approach improves overall detection accuracy and effectively addresses the unique challenges presented by the dynamic nature of cyber threats in EV infrastructures.

**Table 2: CNN-LSTM Model Parameters**

| Layer    | Type       | Parameter    | Value | Output Shape                   |
|----------|------------|--------------|-------|--------------------------------|
| Conv1D   | Layer      | in_channels  | 1     | (batch_size, 16, seq_length-1) |
|          |            | out_channels | 16    |                                |
|          |            | kernel_size  | 2     |                                |
|          |            | stride       | 1     |                                |
| LSTM     | Layer      | input_size   | 16    | (batch_size, 1, 64)            |
|          |            | hidden_size  | 64    |                                |
|          |            | num_layers   | 1     |                                |
|          |            | batch_first  | True  |                                |
| Linear 1 | Layer      | in_features  | 64    | (batch_size, 128)              |
|          |            | out_features | 128   |                                |
| ReLU     | Activation | -            | -     | (batch_size, 128)              |
| Linear 2 | Layer      | in_features  | 128   | (batch_size, 3)                |
|          |            | out_features | 3     |                                |

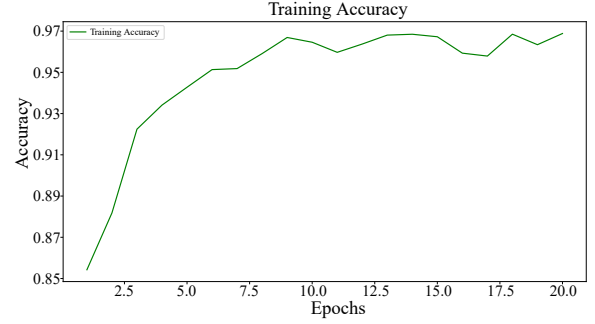
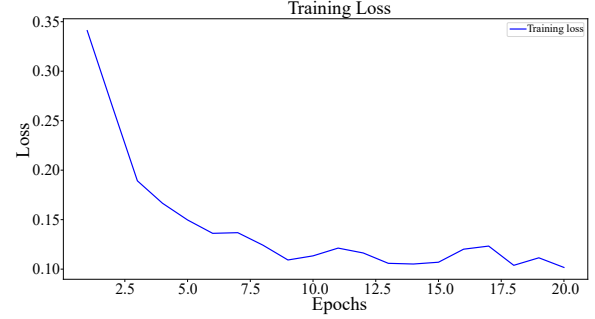
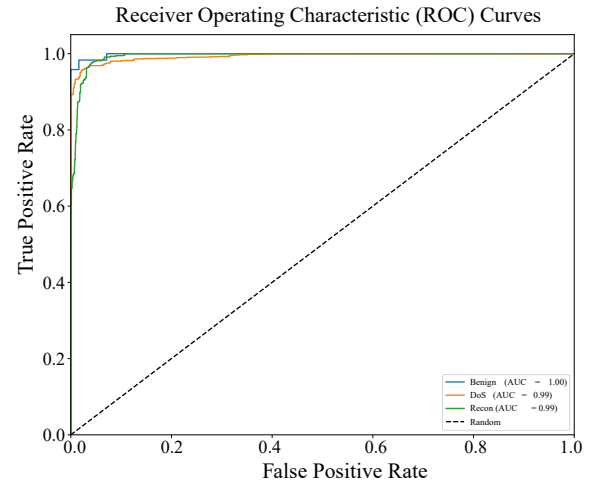
The implementation followed a structured approach, emphasizing model parameters. We trained the CNN-LSTM model for a total of 20 epochs, allowing sufficient time for the model to learn complex patterns in the data. We set the batch size to 32 to optimize memory usage and ensure stable convergence during training. The layers of the model were carefully configured, as detailed in Table 2. The initial layer consisted of a 1D convolutional layer that processed the input data, followed by an LSTM layer designed to maintain information over time. Two linear layers were included for classification, where activation functions were applied to introduce non-linearity.

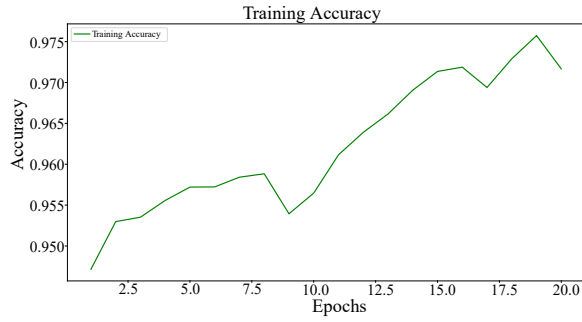
The performance of the model was evaluated on the basis of several key metrics, summarized in Table 3. Both the DNN-CNN-LSTM and CNN-LSTM models achieved high accuracy rates, with the CNN-LSTM model slightly outperforming the DNN-CNN-LSTM model across all evaluated metrics. Specifically, the CNN-LSTM model attained an accuracy of 0.9752, precision of 0.9755, recall of 0.9752, and an F1 score of 0.9753. Moreover, the area under the receiver operating characteristic curve (AUC) was particularly noteworthy, reaching a value of 0.9989, indicating an excellent ability to distinguish between benign and malicious activity. Training accuracy, loss, and ROC curve for EVSE-A and EVSE-B are illustrated in figure 2 and 3 respectively.

These findings underscore the effectiveness of the hybrid model in detecting cyber attacks in EVCS, illustrating its potential applicability in real-world scenarios. The superior performance of the CNN-LSTM model suggests that leveraging CNN for spatial

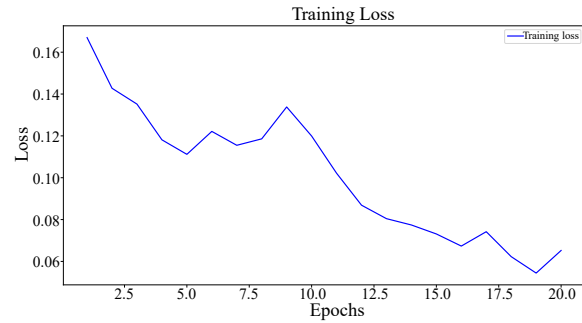
**Table 3: CNN-LSTM Model Performance Metrics**

| Metric    | EVSE-A | EVSE-B | Combined |
|-----------|--------|--------|----------|
| Accuracy  | 0.9715 | 0.9752 | 0.9754   |
| Precision | 0.9719 | 0.9755 | 0.9757   |
| Recall    | 0.9715 | 0.9752 | 0.9754   |
| F1 Score  | 0.9715 | 0.9753 | 0.9754   |
| AUC       | 0.9714 | 0.9989 | 0.9851   |

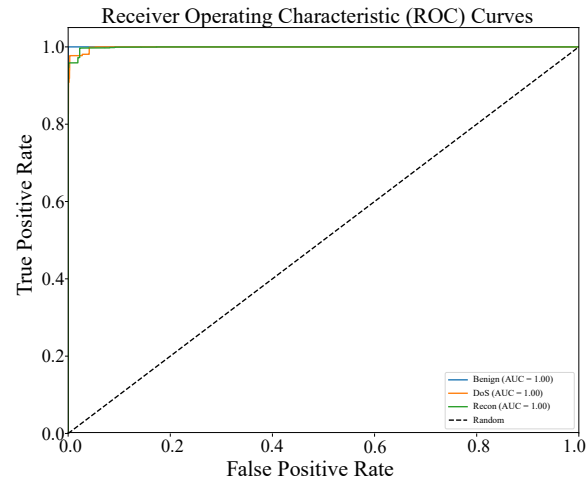

**(a) Training Accuracy for EVSE-A Testbed**

**(b) Training Loss for EVSE-A Testbed**

**(c) ROC Curves for EVSE-A Testbed**
**Figure 2: Performance metrics of our model on the EVSE-A testbed**



(a) Training Accuracy for EVSE-B Testbed



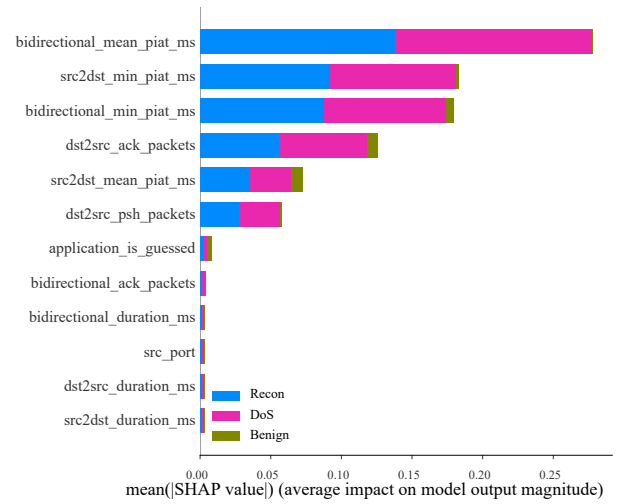
(b) Training Loss for EVSE-B Testbed



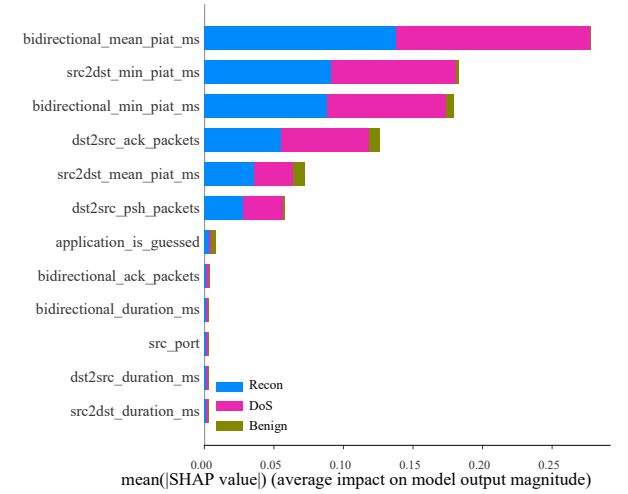
(c) ROC Curves for EVSE-B Testbed

**Figure 3: Performance metrics of our model on the EVSE-B testbed**

feature extraction and LSTM for temporal sequence learning can significantly enhance the robustness of cybersecurity measures in critical infrastructure.



(a) Shap result for EVSE-A Testbed



(b) Shap result for EVSE-B Testbed

**Figure 4: Performance metrics of our model on the EVSE-B testbed**

Finally, we utilized SHAP to assess the impact of these features on the performance of our final CNN-LSTM model. Our results indicated that all 12 features play a significant role in detecting benign, DoS, and reconnaissance data. However, 6 features—namely, `bidirectional_mean_piat_ms`, `src2dst_min_piat_ms`, `bidirectional_min_piat_ms`, `dst2src_ack_packets`, `src2dst_mean_piat_ms`, and `dst2src_psh_packets`—exhibit a particularly strong influence on class detection. Figure 4

## 5 Conclusion

In this study, we introduced a deep learning-based anomaly detection system specifically designed for the EVCS, leveraging the CICEVSE2024. Our hybrid architecture, which integrates CNN and LSTM networks, exhibited exceptional performance in identifying a wide range of cyber threats within the EV charging ecosystem.



The evaluation of our model on two CSs yielded accuracy rates of 97.15% and 97.52%, with precision scores of 97.19% and 97.55%, and recall scores of 97.15% and 97.52%. The F1 scores were 97.15% and 97.53%, while the AUC reached 97.14% and 99.89% for the two stations, respectively. These metrics underscore the model's effectiveness in differentiating between normal operations and potential cyber attacks, highlighting the critical need for enhanced security measures in EVCS. By employing both temporal and spatial feature extraction, our hybrid approach significantly improves the system's reliability in real-world applications, effectively addressing the unique challenges associated with the dynamic nature of network traffic in EV charging systems. After applying XAI using SHAP, we identified a significant impact of our selected 12 features in distinguishing between benign and attack data. These features can be used in further research to detect benign and attack data using DL methods.

The insights gained from this research make a substantial contribution to ongoing initiatives aimed at strengthening cybersecurity within the EV ecosystem. Looking ahead, future work will concentrate on broadening the dataset to include a wider variety of attack scenarios and optimizing the model for real-time deployment. By enhancing the resilience of EV charging infrastructure against cyber threats, we aspire to increase user confidence and encourage the secure adoption of electric vehicles.

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